

REDUNDANT COVERAGE OF ELECTRODE ARRAYS UNDER REAL CONTACT FAILURE: A TOPOLOGICAL CERTIFICATE, AND WHAT IT DOES NOT PREDICT

ADITI BOSE, DEEP BHATTACHARJEE, AND USHASHI BHATTACHARYA

ABSTRACT. Electrode arrays lose contacts, so the region an array can be trusted to record is the region its contacts see at least k times, not the area they see once. We compute the topology of that region from which contact footprints overlap, by Sheehy’s multicover nerve theorem, and test it on the channel failures marked by data curators in 153 recordings from a 64-channel EEG cap. On every real failure pattern, the certificate equals the exact topology of the k -fold region. Real failures are not random. Failed channels are neighbours more often than chance, failure sets recur within a subject, and a minority of recordings lose far more of the target than random failures would take, although the pre-specified primary test of uncovered area did not reach significance ($p = 0.068$). The clustering survives a null that keeps every channel’s own failure rate, and neighbouring channels fail together over a correlation length of about 36° . Under a plan fixed before any signal was loaded, the redundancy left around a channel predicted how well its signal could be rebuilt from the other channels. All four pre-specified tests agreed, and the effect survives removing recording-level effects. But redundancy depth is a sum of pairwise distance terms, and out of sample the topology of the surviving coverage changed the explained variance by at most 0.006 beyond the distances to the nearest survivors. An array optimised for double coverage did not survive the real failures better than the standard cap. The certificate is exact and useful for auditing an array. On this evidence, its physical content is geometric.

1. INTRODUCTION

An electrode array on the scalp or the cortex is judged by what it sees, and it keeps being judged after some of its contacts fail. A contact can lose its seal, pick up line noise, or sit over a region it cannot record from. In the first sessions of the EEG dataset studied here, the curators discarded a median of 4 of 64 channels per recording and up to 31, and only 12 of 111 recordings kept all 64. What matters for an array is therefore not the area its contacts see once, but the region they see at least k times, and whether that region still contains the target when $k - 1$ of them are gone.

This paper asks three questions of that region, on real data. Can its shape be certified from which contact footprints overlap, without rasterising anything? Do real contact failures damage it the way random failures would? And does redundancy measured this way predict something physical: how well the signal of a lost channel can be rebuilt from the channels that remain?

The first question has a known answer. For a family of convex footprints, the region seen at least k times is homotopy equivalent to a simplicial complex built from the nerve of the family: Sheehy’s multicover nerve theorem [16]; see also [6, 3] and, for the case $k = 1$ and the use of such complexes for sensor coverage, [1, 5]. We use it as a tool and claim no new theorem about it. What is new here is empirical: the certificate is run on the failure patterns of 153 real EEG recordings, where the curators of an open dataset marked the bad channels [9, 10]; its answers are checked against the exact topology of the arrangement of cap boundaries; and the redundancy it measures is tested, under an analysis plan fixed before any signal was loaded, against the quality with which each channel is rebuilt from the others.

The answers are mixed, and we report them as they came out. The certificate agrees with the exact topology on every real failure pattern. Real failures are not random: they fall on neighbouring

channels, they recur in the same subject, and a minority of recordings lose far more of the target than random failures of the same size would take. The clustering is not a gradient in failure rate. It survives a null that keeps every channel's failure rate, and channels fail together over a correlation length of about 36° . Under a plan fixed before any signal was loaded, the redundancy that failures leave around a channel predicts how well that channel's signal is rebuilt from the others. But the redundancy that predicts is a sum of distance terms (Proposition 4), and out of sample the topology of the surviving coverage adds nothing measurable to the distances to the nearest survivors. An array optimised for tight double coverage did not survive the real failures better than the standard cap.

2. THE CERTIFICATE

Let X be the scalp sphere or the cortical surface with its geodesic metric, $T \subset X$ the target, and $B_1, \dots, B_n$ the footprints of n contacts, geodesic balls of a common radius θ about the contact positions $c_1, \dots, c_n$. The multiplicity of a point is $m(x) = \#\{i : x \in B_i\}$ and the k -fold region is $R_k = \{x : m(x) \geq k\}$, so that $R_1 \supseteq R_2 \supseteq \dots$.

Definition 1. The nerve N of the family is the simplicial complex of index sets σ with $\bigcap_{i \in \sigma} B_i \neq \emptyset$. Its k -shadow $\Delta_k(N)$ has one vertex for each face of N with exactly k elements, and a set of such vertices spans a simplex when the union of their index sets is a face of N .

Theorem 2 (Sheehy [16]). *If every nonempty intersection of footprints is contractible (for instance, if the footprints are convex), then $\Delta_k(N)$ is homotopy equivalent to R_k for every $k \geq 1$.*

It is the nerve theorem [1] applied to the cover of R_k by the sets $B_\sigma = \bigcap_{i \in \sigma} B_i$ with $|\sigma| = k$: a family of such sets meets exactly when the union of their index sets is a face of N , so the nerve of that cover is $\Delta_k(N)$, and each of its intersections is itself an intersection of footprints.

Everything about the shape of R_k can thus be read from which footprints meet, and for equal balls in the plane, or caps of radius below 30° on the sphere, Helly's theorem [11] reduces that to pairs and triples. We call the *certified level* of an array the largest k such that $\Delta_j(N)$ has Betti numbers $(b_0, b_1) = (1, 0)$ for every $j \leq k$. The condition is not monotone in k and has to be checked at every level.

The certificate gives the shape of R_k , not its position. It neither proves nor refutes that R_k contains the target. A second component or a hole can lie outside T while $T \subseteq R_k$, and a disk-like R_k can miss part of T ; Section 3 has an example of each. Whether T is covered is therefore always checked directly here. A combinatorial condition, checkable on N and a description of T , that forces $T \subseteq R_k$ is open, and is the first of the two problems of Section 6.

The certificate does give a guarantee that is purely combinatorial.

Proposition 3 (dropout). *Under the hypothesis of Theorem 2, suppose $\Delta_{q+1}(N)$ is connected and every contact belongs to a face of N with $q+1$ elements. Then for any set F of at most q failed contacts, the union of the surviving footprints is connected.*

Proof. A point of R_{q+1} lies in $q+1$ footprints, at most q of which failed, so $R_{q+1} \subseteq \bigcup_{i \notin F} B_i$. By Theorem 2 R_{q+1} is connected. A surviving footprint B_i contains a point of a $(q+1)$ -fold intersection, hence a point of R_{q+1} , and is itself connected. The union of the surviving footprints is R_{q+1} together with connected sets that each meet it, so it is connected. $\square$

We call the largest such q the *dropout margin*. One more observation turned out to decide how the signal-level test of Section 5 can be read, so we state it here.

Proposition 4 (averaged multiplicity is pairwise). *Let μ be the area measure, c a contact, and S a set of other contacts. The mean multiplicity of the surviving footprints over the footprint of c ,*

$$\lambda(c, S) = \frac{1}{\mu(B_c)} \int_{B_c} \sum_{j \in S} \mathbf{1}_{B_j} d\mu = \sum_{j \in S} \frac{\mu(B_c \cap B_j)}{\mu(B_c)},$$

is additive over S . On the sphere, with caps of common radius θ , each term depends only on the angle γ_{cj} between c and j : $\lambda(c, S) = \sum_{j \in S} K_\theta(\gamma_{cj})$, where $K_\theta(\gamma)$ is the fraction of a cap covered by a second cap at angle γ , which vanishes for $\gamma \geq 2\theta$.

Proof. Exchange the finite sum and the integral. On the sphere, the rotation group acts transitively on pairs of points at a given angle, so $\mu(B_c \cap B_j)$ depends on γ_{cj} alone. $\square$

So λ , the natural “redundancy depth” of a contact, is a kernel sum over the distances from c to the survivors, one survivor at a time. It cannot see how the survivors overlap one another, and that overlap is exactly what the topology of R_k records. The private territory $\pi(c, S) = \mu(B_c \setminus \bigcup_{j \in S} B_j) / \mu(B_c)$ does depend on it, through inclusion and exclusion over all orders, and so does every Betti number. A test in which λ predicts a physical quantity is therefore a test of geometry, however it comes out. Only a quantity that is not additive over survivors can show that topology does work that distances do not.

3. PLACEMENT ON A TEMPLATE CORTEX

On a 3777 mm² patch of the left parietal pial surface of the colin27 template [12, 14], with footprints of geodesic radius 8 mm, the 8 × 8 grid of the FieldTrip tutorial layout, transferred to the template, leaves 2082 mm² (55.1%) of the patch unseen, with $\Delta_1(N)$ of Betti numbers (9, 2). The same number of contacts placed by farthest-point sampling in the geodesic metric leaves 1.0 mm² unseen, with Betti numbers (1, 1), and 66 contacts leave nothing unseen at certified level 1 (Fig. 1). The comparison is between placements, not devices. The grid was laid out for a larger region, and the farthest-point design uses any pial vertex, including sulcal depths that a subdural grid cannot reach. The constraint that binds a real subdural array is the *crown floor*: with all 3016 gyral-crown vertices of the patch as contacts, some point of the patch is still 12.22 mm from every one of them. So no surface array whose footprint radius is below 12.22 mm can see the whole patch, however many contacts it has.

Geodesic footprints on a folded surface are not convex, so Theorem 2 is not guaranteed there, and every certificate on the cortex was checked against the k -fold region computed directly on the mesh. The check matters. With 112 contacts at 9 mm, the whole patch is seen at least twice by direct computation (0.0 mm² seen fewer than twice), and the certificate gives $\Delta_2(N)$ Betti numbers (1, 0). But the mesh finds (1, 1) for R_2 : a hole outside the patch that the nerve misses, with 961 of the 16835 nerve faces failing the discrete disk test. With 96 contacts at 10 mm the two computations agree at every level and the certified level is 2, yet 1.7 mm² of the patch is seen only once. The certificate is correct there and still does not imply containment.

4. REAL CONTACT FAILURES

Data and plan. OpenNeuro ds003775 v1.2.1 [10] holds resting-state EEG from 111 adults on a 64-channel BioSemi cap, 42 of them recorded again 2–3 months later, with the channels that the curators’ automated pipeline marked bad in each of the 153 recordings [9]. The data descriptor gives two channel retention figures, and the failure table rebuilt from the dataset’s public mirror reproduces both exactly. Each channel gets a spherical-cap footprint. The target T is the scalp inside the Fpz–T7–Oz–T8 circumference, and the intact cap sees all of T once, twice and three times from cap radii 16.0, 23.0 and 25.3 degrees. The analysis plan was fixed before any outcome was computed. It set the primary radius at 23.5°, where the intact cap sees T twice, so that no single failure can uncover any of it; the first sessions as the primary cohort and the second as replication; and, as the null, each failure set replaced by a uniformly random set of the same size.

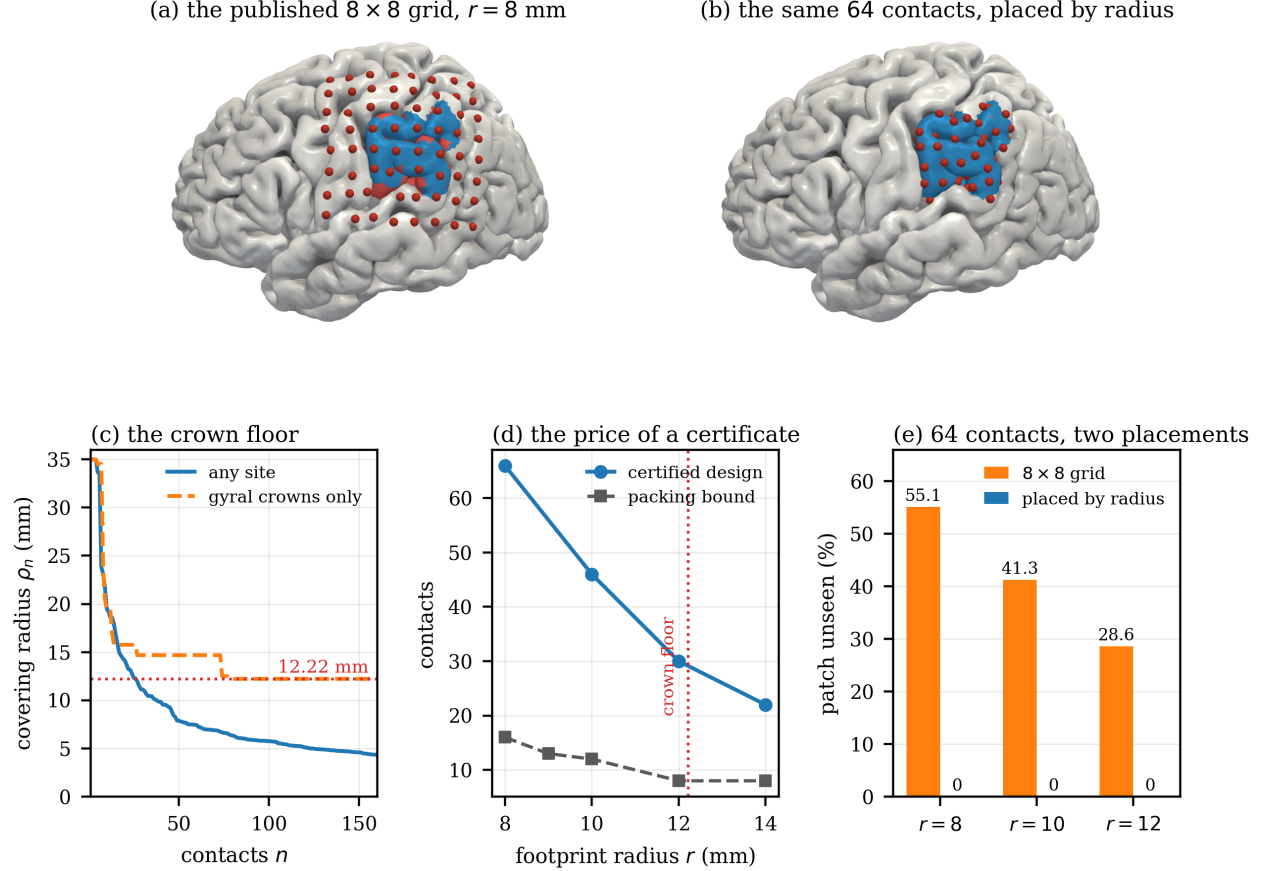


FIGURE 1. Placement on the colin27 template. (a) The 8×8 tutorial grid and (b) 64 contacts placed by geodesic farthest-point sampling, with the target patch in blue and 8 mm footprints. (c) Covering radius against contact count, for sites anywhere on the patch and for gyral crowns only; the crown curve cannot fall below 12.22 mm. (d) The smallest contact count with certified level 1 at each footprint radius, against the packing lower bound. (e) Patch unseen by the two 64-contact placements.

The certificate on real failures. On all 153 surviving caps the Betti numbers of Δ_1 and Δ_2 , computed from intersection data alone, equal those of the k -fold region computed exactly from the arrangement of boundary circles, and they agree on 1000 of 1000 random failure sets. A raster of the multiplicity function agrees on 140 of the 153; in the others it reports extra components or holes, which is a resolution effect.

Pre-specified tests. The primary outcome was the uncovered fraction of T after failure. Its one-sided Wilcoxon signed-rank test against the random expectation, over the 99 first-session recordings with at least one failure, gave $p = 0.068$, so the primary test *failed* at $\alpha = 0.05$. The plan named a second test for the same outcome, a Monte Carlo test of the cohort total, and it gave $p = 1.0 \times 10^{-4}$: real failures uncover 3.0 times the area that random failures of the same sizes uncover (mean 1.05% of T against 0.35%). The two disagree because the effect is carried by a minority of recordings. 32 recordings lose more than their random expectation and 54 lose less, but those that lose more lose much more. Because the plan did not say how to combine the two tests, we report the primary endpoint as not established.

The secondary tests were clearer. Failed channels are neighbours more often than random sets of the same size (1455 adjacent failed pairs against 1134 expected, $p = 4.1 \times 10^{-7}$; replication

$p = 0.004$). Channels fail at very different rates, from 41% for P9 down to none (chi-square 367 against a null mean of 49), and the rate rises towards the rim (Spearman 0.82 with colatitude). Failure sets recur within a subject (mean Jaccard index 0.100 between sessions against 0.057 for random re-pairings, $p = 1.0 \times 10^{-3}$). The descriptor’s quality rule, “more than 90% of channels kept”, disagrees with the coverage outcome in 6 recordings that pass it and lose target, and 6 that fail it and keep all of T . Real failures also punch fewer holes into the surviving coverage than random ones (mean b_1 0.24 against 0.68, $p = 3.4 \times 10^{-8}$). Clustered failures take out a region at the edge instead of scattering gaps through the interior, and that difference is topological.

Clustering beyond a gradient in failure rate (exploratory). The null of the plan draws failures uniformly, but channels do not fail uniformly. The channels that fail most are rim channels, and they are each other’s neighbours, so a rate gradient alone would produce adjacent failures and rim losses with no interaction between failures at all. We therefore compared the real failures with null cohorts that keep each recording’s number of failures *and* each channel’s number of failures, sampled by curveball swaps [17]. The clustering survives. There are 1455 adjacent failed pairs against 1114 (95% null range 1074–1154). The mean uncovered target is 1.05% against 0.25% (0.16–0.35%). The reach needed at 95% is 46.0° against 30.7° (28.3–33.5%). In each case no null cohort of 1000 reached the observed value ($p \leq 0.001$), and the second sessions give the same answer. Channels at $10\text{--}20^\circ$ from each other fail together 1.72 times as often as their own rates predict (second sessions 1.84; the 95% null range reaches 1.10). The excess falls to $1/e$ of that by $\xi \approx 36^\circ$ (bootstrap 95% interval $20\text{--}45^\circ$; second sessions 34° , $17\text{--}91^\circ$), about two inter-electrode spacings of 18° (Fig. 3c). Failure on this cap is a spatially correlated process with a measurable correlation length. Its source is not identified here; a shared electrolyte bridge, a cap that sits poorly over one region, and muscle or movement artefact over the temporal rim are all candidates.

The reach real failures demand (exploratory). For each recording, $\rho_1(S)$ is the smallest cap radius at which the survivors see all of T . Keeping all of T seen in 95% of first-session recordings takes 46.0° , against 30.9° (95% interval $28.1\text{--}33.6^\circ$) for random failures of the same sizes. At 90% it takes 37.6° against 27.3° .

A layout optimised for double coverage (exploratory). Minimising the 2-fold covering radius of T over 64 positions brings it from 23.0° to 19.8° . Carrying the real failures to the new layout by one assumption (a contact fails where the nearest cap channel failed), the optimised layout lost more of T than the standard cap in 34 recordings and less in 17 ($p = 1.00$ for improvement). Restricting the optimisation to a colatitude chosen on one half of the subjects and scored on the other did not help in either fold ($p = 0.34$ and 0.90). Tight double coverage protects against scattered losses, and the real losses are clustered at the rim.

5. DOES REDUNDANCY PREDICT HOW WELL A LOST CHANNEL IS REBUILT?

Coverage is a statement about footprints, and a footprint is a model. The test of whether it corresponds to anything physical is the signal. If a channel is lost, its signal can be estimated from the channels that remain, and the estimate should be better where the loss left more redundancy behind. We fixed the plan for this test before any signal was loaded (Methods). Every good channel of every recording was removed in turn and rebuilt from all other good channels by spherical-spline interpolation [15], and the rebuild was scored by the Fisher-transformed correlation z between the true and rebuilt signal over four minutes. That gave 8673 rebuilds in 152 recordings, with median correlation 0.886. One second-session file of the 153 published recordings contains no epochs and could not be used.

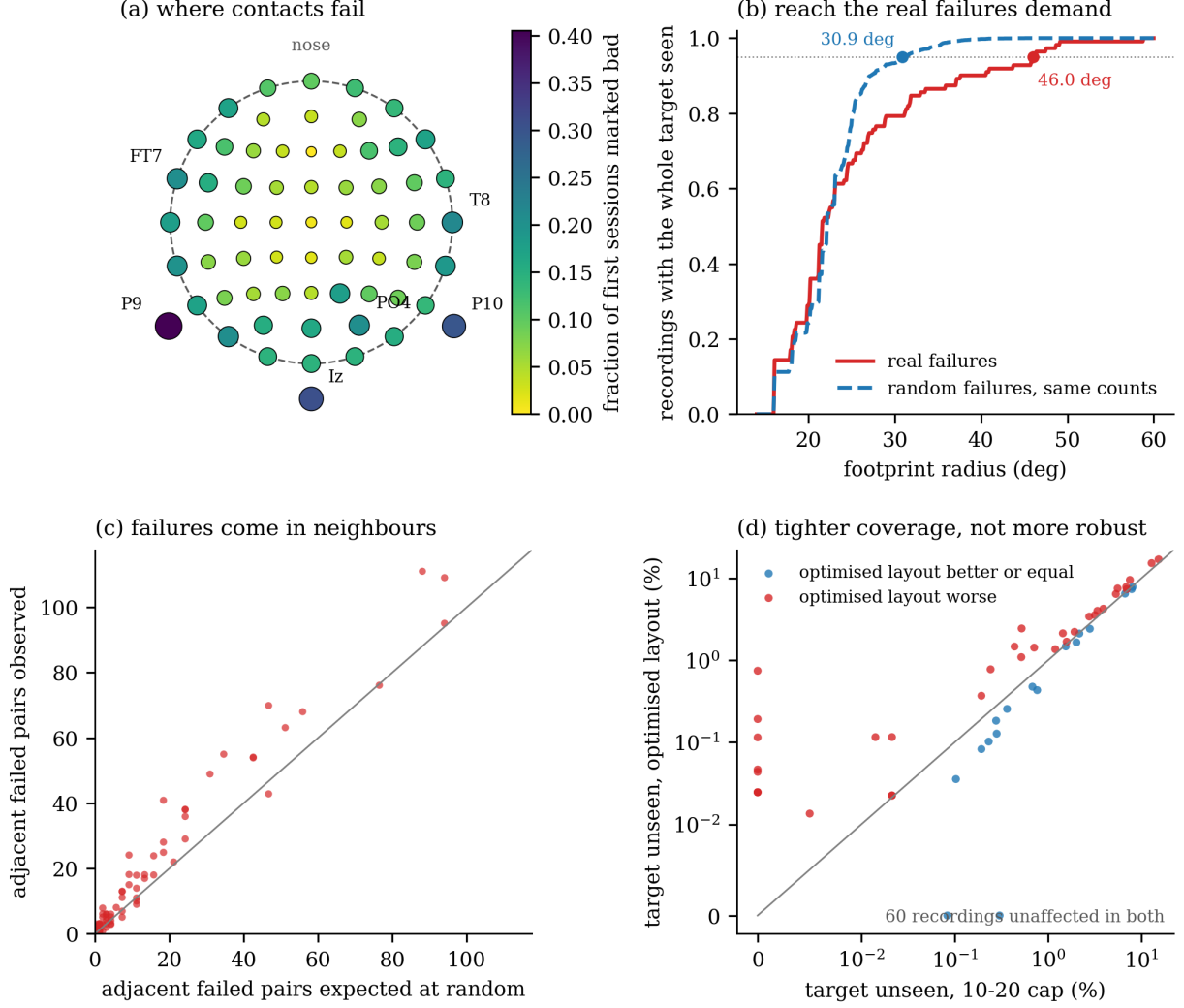


FIGURE 2. Real contact failures on the 64-channel cap, first sessions. (a) Fraction of recordings in which each channel was marked bad. (b) Share of recordings in which the survivors see all of T , against cap radius, for the real failure sets and for random sets of the same sizes. (c) Adjacent failed pairs, observed against the random expectation, one point per recording. (d) Target unseen at 23.5° by the standard cap and by the layout optimised for 2-fold coverage, under the same real failures.

Pre-specified tests. The primary test is a natural experiment inside each channel. A channel's position never changes, and what varies from recording to recording is which of its neighbours failed, and hence its redundancy depth λ . Across the first-session recordings, the Spearman correlation between λ and z was positive for all 64 channels (median 0.198, $p = 1.8 \times 10^{-12}$). Private territory worked in the opposite direction, as specified (62 of 64 channels negative, median -0.158 , $p = 6.0 \times 10^{-12}$). Across channels within a recording, λ still predicted z given the distance d_1 to the nearest survivor (median partial correlation 0.092, positive in 75 of 111 recordings, $p = 7.7 \times 10^{-7}$). In the second sessions the primary test replicated (median 0.294, 60 of 64 channels positive, $p = 3.3 \times 10^{-12}$). All four pre-specified tests came out in the planned direction.

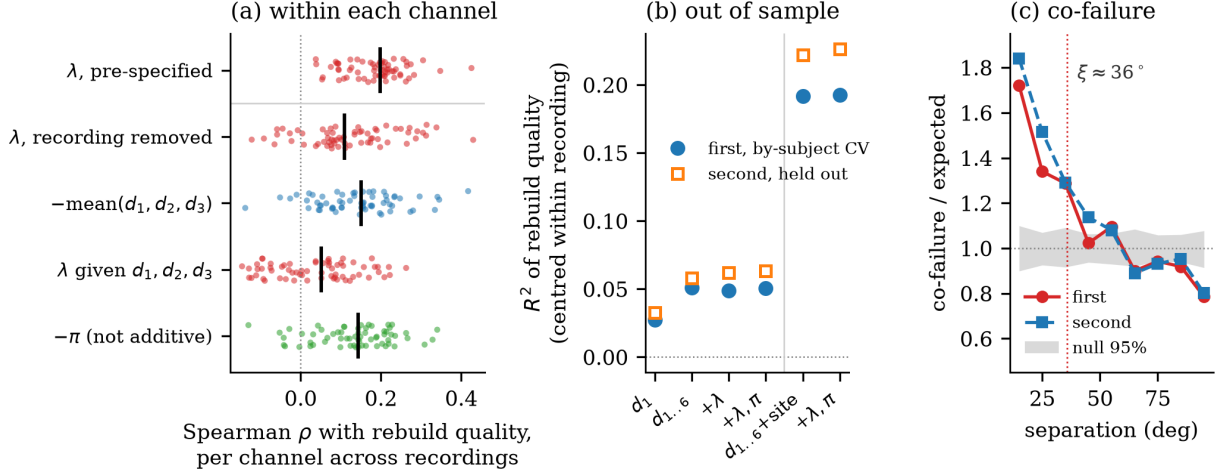


FIGURE 3. (a) Within-channel Spearman correlation between each predictor and rebuild quality, one point per channel, first sessions, with the median marked. The top row is the pre-specified test. Those below are exploratory, with rebuild quality centred within each recording. (b) Out-of-sample R^2 of rebuild quality centred within recording, for nested models: the nearest distance; the six nearest distances; with λ ; with λ and π ; the six distances with the channel site; and with λ and π added to that. (c) Co-failure of channel pairs against their angular separation, observed over expected under a null that keeps every recording’s failure count and every channel’s failure rate, with the 95% range of that null (first sessions) and the correlation length ξ at which the excess falls to $1/e$.

What the result can and cannot say (exploratory). Three analyses written after these tests decide how to read them.

First, the within-channel design does not control for the recording. A recording with many failed channels gives all of its channels a low λ , and if such recordings are also noisier, every rebuild in them is worse, with no local effect at all. With z centred within each recording, the within-channel correlation with λ falls from 0.198 to 0.109 but stays positive in 55 of 64 channels ($p = 3.2 \times 10^{-10}$; second sessions 0.161, $p = 3.6 \times 10^{-9}$). Local redundancy is real, and it predicts how well a lost channel can be rebuilt.

Second, by Proposition 4, λ is a sum over survivors of a function of each survivor’s distance. On all 8673 rows it equals $\sum_j K_\theta(\gamma_{ej})$ to within 0.0022, which is the sampling error of the footprint areas. It is a geometric quantity. A purely metric predictor, the mean distance to the three nearest survivors, does at least as well within channel (median 0.151 with z centred, $p = 2.4 \times 10^{-11}$), and given those three distances, what λ adds shrinks to a median of 0.052 (positive in 39 of 64 channels, $p = 0.008$).

Third, prediction out of sample. With z centred within recording, the six nearest distances explain $R^2 = 0.051$ under leave-one-subject-out cross-validation (held-out second sessions 0.058). Adding λ and the non-additive private territory π changes R^2 by at most 0.006 in either evaluation, with or without the channel site in the model. The channel site itself adds 0.14 (Fig. 3b). So redundancy predicts how well a lost channel is rebuilt, but the part of it that predicts is carried by distances. The topology of the surviving coverage adds nothing measurable here.

6. DISCUSSION

What the evidence supports. The certificate is exact where its hypothesis holds: on the scalp it reproduced the exact topology of the k -fold region for every real failure pattern. On a folded cortex, where geodesic footprints are not convex, it can miss a hole, so it has to be checked against a direct computation there. It gives the shape of the k -fold region and not whether that region contains the target. Real failures are clustered, recur within a subject and concentrate at the rim, and they are correlated in space beyond what each channel’s failure rate explains, over about 36° . Redundancy predicts how well a lost channel is rebuilt, but through distances. What has not been shown is that the topology of the coverage changes an outcome that distances do not already predict, and that is the claim a physical or clinical reader would need. This paper does not make it.

What would change this picture. Four results would. The first is clinical. Whether the certified coverage of the resected territory predicts seizure freedom after epilepsy surgery can be tested on open intracranial recordings that carry electrode coordinates, resection information and surgical outcome, such as OpenNeuro ds003029 [13]. The protocol reconstructs each patient’s pial surface, places that patient’s own contacts, computes the covering radius and certified level over the resected region, and tests them against outcome, all under a plan fixed before any outcome label is read. Incomplete resection of the seizure onset zone has been reported not to predict outcome [7], so the result could go either way, and a null would end the clinical claim. Before any such dataset is used, its data descriptor and the independent studies that reuse it should be checked, to make sure the released version is the one that produced published results.

The second is a law for correlated failure. The required reach at 95% is set by the largest region that the failures empty. A theory giving that reach from a measured failure correlation length, in the manner of correlated site removal on a covering, would predict the reach before the data arrive. It would have to be confirmed on an independent dataset with curated channel status. The correlation length measured here is the input such a theory needs, but it is not yet the theory. Its interval is wide, and it comes from one cap, one laboratory and one curation pipeline.

The third is a design that wins out of sample. The layout that lost was optimised for tight double coverage. A layout optimised against the loss of adjacent pairs at the rim, chosen on one dataset and scored on another, would turn the negative result of Section 4 into a positive one.

The fourth is mathematical. Two problems are open: a combinatorial condition, checkable on N and a description of T , that forces $T \subseteq R_k$; and a stability theorem for the two-parameter module indexed by footprint radius and multiplicity [2] under perturbation and deletion of contacts, extending the one-parameter bounds [4] measured in the supplementary code.

7. METHODS

Footprints and targets. Scalp positions are the 64 BioSemi positions of MNE-Python’s standard montage [8] as unit vectors, with Cz at the pole and the Fpz–T7–Oz–T8 circumference at colatitude 92° . Areas on the sphere are measured on a Fibonacci lattice of 4×10^6 points, of which about 2.07×10^6 fall in T . The cortical surface is the left pial surface of colin27 from the FieldTrip toolbox, 175 409 vertices, with every edge in two triangles and Euler characteristic 2. Geodesic distances are Dijkstra distances on the mesh edges, and a triangle belongs to a footprint when all three of its vertices do.

Nerves and Betti numbers. For caps of radius below 30° , any subfamily whose triples meet lies in an open hemisphere about each of its centres, and the gnomonic chart maps it to planar convex sets, so Helly’s theorem makes the nerve the flag completion of its triangles. A triple of caps meets when the minimal enclosing cap of its centres has radius below θ . Betti numbers b_0, b_1 of $\Delta_k(N)$ are computed over $\text{GF}(2)$ from its 2-skeleton. The exact reference for R_k on the sphere is computed

from the arrangement of the boundary circles, with no nerve involved; the raster reference counts components and holes of $\{m \geq k\}$ in a stereographic chart of 4000^2 pixels.

Statistics. All tests are one-sided in the direction stated in the plans, with $\alpha = 0.05$, and use the Wilcoxon signed-rank test [18] or Monte Carlo tests with $p = (1 + \#\{\text{null} \geq \text{observed}\})/(1 + M)$. Seeds are fixed in the code. The plans for the failure analysis and the signal-level test are the docstrings of `run_eeg_failures.py` and `run_signal_rebuild.py`, committed before the respective outcomes were computed. Every analysis labelled exploratory was written afterwards.

Signal-level test. The curated epochs of ds003775 v1.2.1 (band-pass 1–45 Hz, average reference, 60 epochs of 4 s at 1024 Hz per recording) were fetched by S3 object version and checked against the size and MD5 of each file’s git-annex key. Channels the curators marked bad had been interpolated by their pipeline and were used neither as targets nor as inputs. In each recording, each good channel c was removed and rebuilt from all other good channels by spherical-spline interpolation [15] (stiffness 4, 50 Legendre terms, regularisation 10^{-5} , the MNE-Python defaults [8]). Rebuild quality is the Fisher transform $z = \text{artanh } r$ of the correlation r between the true and rebuilt signal over all 240 s. For the survivors S and the primary radius 23.5° we recorded $\lambda(c, S)$, $\pi(c, S)$, the number of survivors within 23.5° and the angle d_1 to the nearest survivor. The pre-specified primary test (P) takes, for each channel that is a good target in at least 10 first-session recordings with at least 3 distinct values of λ , the Spearman correlation between λ and z across those recordings, and tests with a one-sided Wilcoxon signed-rank test over channels that these correlations are positive. A channel’s position never changes, so what varies between recordings is which of its neighbours failed: a natural experiment inside each channel. Test S1 is the same with π in the negative direction; S2 is the partial Spearman correlation of z with λ given d_1 across channels within each first-session recording, one-sided over recordings; S3 repeats P on the second sessions.

Exploratory analyses. (i) Proposition 4 was checked on every row by evaluating K_θ by quadrature. (ii) The within-channel test was repeated with metric predictors and with partial correlations given the nearest one and three distances. (iii) Rebuild quality, centred within each recording to remove every recording-level effect, was predicted by nested least-squares models: d_1 and d_1^2 (M1); the six nearest distances and their squares (M2); M2 with λ, λ^2 (M3); M3 with π and $\mathbf{1}[\pi > 0]$ (M4); M2 with 63 channel indicators (M5); M5 with λ, π (M6). Models were scored by leave-one-subject-out cross-validated R^2 on the first sessions, and, fitted on all first sessions, on the second sessions as a held out set. (iv) The clustering statistics were compared with a null that keeps each recording’s number of failures and each channel’s number of failures, sampled by curveball swaps [17] (1000 null cohorts, 1000 swaps apart after a burn-in of 20 000), and with the uniform null of the plan. (v) Pair correlation of failure: for channel pairs binned by angular separation in 10° bins, the number of recordings in which both failed, divided by its mean over the same null cohorts. The correlation length is the separation at which the excess over one falls to $1/e$ of its value in the first occupied bin, interpolated linearly between bin centres, with a 95% interval from 2000 bootstrap resamples of recordings.

DATA AVAILABILITY

OpenNeuro ds003775 v1.2.1 (doi:10.18112/openneuro.ds003775.v1.2.1, CC0). The cleaned-epoch files are fetched by their S3 object versions, and the version ids are recorded with each file’s size and MD5 in `data/ds003775_epochs_manifest.tsv`, since later versions of the dataset dropped them. The colin27 surface is derived from FieldTrip commit 8e2307d. No patient data are used.

CODE AVAILABILITY

All code, the analysis plans and every result file quoted here are at <https://github.com/creelie/kshadow-coverage>. Every number in this manuscript is a macro generated from those result files by `make_paper_numbers.py`.

AUTHOR CONTRIBUTIONS

[To be completed by the authors.]

COMPETING INTERESTS

The authors declare no competing interests.

REFERENCES

1. Karol Borsuk, *On the imbedding of systems of compacta in simplicial complexes*, Fundamenta Mathematicae **35** (1948), 217–234.
2. Gunnar Carlsson and Afra Zomorodian, *The theory of multidimensional persistence*, Discrete & Computational Geometry **42** (2009), 71–93.
3. Nicholas J. Cavanna, Kirk P. Gardner, and Donald R. Sheehy, *When and why the topological coverage criterion works*, Proceedings of the 28th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2017.
4. David Cohen-Steiner, Herbert Edelsbrunner, and John Harer, *Stability of persistence diagrams*, Discrete & Computational Geometry **37** (2007), 103–120.
5. Vin de Silva and Robert Ghrist, *Coverage in sensor networks via persistent homology*, Algebraic & Geometric Topology **7** (2007), 339–358.
6. Herbert Edelsbrunner and Georg Osang, *The multi-cover persistence of Euclidean balls*, Discrete & Computational Geometry **65** (2021), 1296–1313.
7. Sarah J. Gascoigne et al., *Incomplete resection of the intracranial electroencephalographic seizure onset zone is not associated with postsurgical outcomes*, Epilepsia **65** (2024), e163–e169.
8. Alexandre Gramfort, Martin Luessi, Eric Larson, Denis A. Engemann, Daniel Strohmeier, Christian Brodbeck, Roman Goj, Mainak Jas, Teon Brooks, Lauri Parkkonen, and Matti Hämäläinen, *MEG and EEG data analysis with MNE-Python*, Frontiers in Neuroscience **7** (2013), 267.
9. Christoffer Hatlestad-Hall, Trine Waage Rygvold, and Stein Andersson, *BIDS-structured resting-state electroencephalography (EEG) data extracted from an experimental paradigm*, Data in Brief **45** (2022), 108647.
10. ———, *SRM resting-state EEG, OpenNeuro dataset ds003775, version 1.2.1*, 2022.
11. Eduard Helly, *Über Mengen konvexer Körper mit gemeinschaftlichen Punkten*, Jahresbericht der Deutschen Mathematiker-Vereinigung **32** (1923), 175–176.
12. Colin J. Holmes, Rick Hoge, Louis Collins, Roger Woods, Arthur W. Toga, and Alan C. Evans, *Enhancement of MR images using registration for signal averaging*, Journal of Computer Assisted Tomography **22** (1998), 324–333.
13. Adam Li et al., *Neural fragility as an EEG marker of the seizure onset zone*, Nature Neuroscience **24** (2021), 1465–1474.
14. Robert Oostenveld, Pascal Fries, Eric Maris, and Jan-Mathijs Schoffelen, *FieldTrip: open source software for advanced analysis of MEG, EEG, and invasive electrophysiological data*, Computational Intelligence and Neuroscience **2011** (2011), 156869.
15. F. Perrin, J. Pernier, O. Bertrand, and J. F. Echallier, *Spherical splines for scalp potential and current density mapping*, Electroencephalography and Clinical Neurophysiology **72** (1989), 184–187.
16. Donald R. Sheehy, *A multicover nerve for geometric inference*, Proceedings of the 24th Canadian Conference on Computational Geometry (CCCG), 2012.
17. Giovanni Strona, Domenico Nappo, Francesco Boccacci, Simone Fattorini, and Jesus San-Miguel-Ayanz, *A fast and unbiased procedure to randomize ecological binary matrices with fixed row and column totals*, Nature Communications **5** (2014), 4114.
18. Frank Wilcoxon, *Individual comparisons by ranking methods*, Biometrics Bulletin **1** (1945), 80–83.

CENTRE FOR COMPUTATIONAL NATURAL SCIENCES AND BIOINFORMATICS, INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY, HYDERABAD, INDIA

URL: <https://orcid.org/0000-0002-3036-8223>

[AFFILIATION TO BE SUPPLIED BY THE AUTHOR]

URL: <https://orcid.org/0000-0003-0466-750X>

[AFFILIATION TO BE SUPPLIED BY THE AUTHOR]

URL: <https://orcid.org/0009-0002-2254-3914>